

Kinjal Pandey

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EDUCATION

Banasthali Vidyapith

Bachelor of Technology in Computer Science Engineering

Tonk, Rajasthan

Expected Graduation: May 2027

- **Relevant Coursework:** Data Structures & Algorithms, OOPs, Java Programming, Operating Systems, Database Management Systems, Software Engineering, Data Communication & Networks, Microprocessor & Microcontroller

EXPERIENCE

Quantum Computing Research Intern

Scientific Analysis Group, DRDO

New Delhi, India

May 2025 - July 2025

- Architected Qiskit-based reversible quantum arithmetic circuits (3-bit addition/division), reducing circuit depth and decoherence by 25% via efficient Toffoli (CCX) and CNOT gate decompositions.
- Engineered Python simulations for BB84 and COW Quantum Key Distribution (QKD) protocols, developing decoy-state heuristics to harden encryption for 10K+ defense data points against post-quantum threats (Shor's Algorithm).

IoT Strategy and Development Intern

OTAmation Tech Labs

Indore, MP

July 2024 - Oct 2024

- Architected an ESPHome-based smart lock (Wemos D1 Mini, RC522), engineering a custom C++ firmware component to process low-latency, SPI-based RFID telemetry.
- Optimized microcontroller memory stability across 1K+ hardware cycles by tuning C++ heap allocation and utilizing custom compiler build flags (-fno-exceptions, -DUSE_STORE_LOG_STR_IN_FLASH) for crash-free execution.

PROJECTS

ViSiON-46 | TensorFlow.js (COCO-SSD), Tesseract.js, Web Speech API, Google Maps API

March 2025

- Architected a client-side accessibility engine (TensorFlow.js, Tesseract.js), optimizing memory via requestAnimationFrame loops and extracting base64 text using PSM-6 tuning and Regex NLP sanitization.
- Developed a voice-actuated navigation system, engineering $O(1)$ Hash Set deduplication and a 3000ms stateful audio-throttle to prevent overlapping feedback during high-frequency object tracking.

VeritAz | JavaScript (ES6+), Manifest V3, MediaPipe, Tesseract.js, Canvas API

Jan 2026

- Architected a real-time AI forensic Chrome extension (MediaPipe, Canvas API) for Google Meet, engineering a deepfake detection pipeline that utilizes $O(N)$ stride-based array iteration ($i += 16$) to evaluate blue-channel pixel variance with minimal processing latency.
- Developed an asynchronous threat-detection engine utilizing a DOM MutationObserver and $O(1)$ Hash Set deduplication to parse malicious chat scripts, while concurrently tracking biological liveness (rPPG) via a strictly managed 60-frame rolling variance buffer.

F1 - PitWallAI(F1 race replay console) | React, TypeScript, FastAPI, Python, Three.js, WebSockets, FastF1, PyArrow

April 2025

- Architected a real-time F1 telemetry and 3D replay platform using FastAPI, React, and WebSockets, rendering live race-state visualizations via Three.js.
- Engineered optimized data pipelines with Python and PyArrow to process, cache, and serve high-frequency telemetry (speed, RPM, stint strategy) with minimal retrieval latency.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript (ES6+), SQL (MySQL, Oracle).

Frameworks & Libraries: React.js, Node.js, FastAPI, TensorFlow, PyTorch, OpenCV, NumPy, Matplotlib, Qiskit.

Developer Tools: Git, GitHub, MySQL Workbench, Firebase, Deployment Pipelines (Netlify, Vercel), VS Code, PyCharm.

Core Competencies: Edge Computing, REST API Architecture, Embedded Systems Optimization, Quantum Algorithms, Machine Learning Pipelines.

AWARDS & LEADERSHIP

Awards: 1st Place (HackCelestia '26, AIC-HackVerse, Smart India Hackathon '24); Finalist/Top 15 (Myntra HackerRamp '24, GfG Ideathon).

Technical Content Writer

Microsoft Student Chapter-BV

Tonk, RJ

Jan 2024 - Present

Technical Content Strategist

AlgoByte Banasthali Chapter

Tonk, RJ

Oct 2023 - Oct 2024